

SWAN HILL, Australia

WMO: 948430

Lat: 35.3767S

Lon: 143.5417E

Elev: 233

StdP: 14.57

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.8	34.9	18.2	13.9	82.0	22.6	17.1	77.0	27.9	54.9	25.2	54.7	3.6	160	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.1	102.6	65.5	98.1	64.6	94.3	63.9	71.1	85.2	69.2	84.9	67.6	84.8	13.0	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.7	102.6	74.7	65.0	93.2	73.5	62.2	84.3	72.6	35.1	84.2	33.5	83.9	32.1	84.5	93.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.3	22.0	19.2	DB	28.5	109.4	2.3	3.6	26.9	112.0	25.5	114.1	24.2	116.1	22.6	118.7	(4)
(5)			WB	29.5	75.3	2.1	5.5	28.0	79.2	26.8	82.4	25.6	85.5	24.1	89.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.5	76.1	75.1	69.3	61.3	54.6	49.9	48.7	50.3	54.9	60.4	67.0	71.4	(6)
(7)		DBStd	11.46	7.81	7.51	7.04	6.01	4.85	4.08	3.92	4.36	5.80	6.96	7.86	7.61	(7)
(8)		HDD50	193	0	0	0	1	10	50	73	47	11	1	0	0	(8)
(9)		HDD65	2495	4	6	32	142	322	453	506	455	310	180	64	21	(9)
(10)		CDD50	4397	808	703	598	341	154	47	32	58	158	325	510	663	(10)
(11)		CDD65	1224	347	289	166	32	1	0	0	7	39	124	219		(11)
(12)		CDH74	16738	4809	3637	1977	437	18	0	0	6	149	726	1835	3144	(12)
(13)		CDH80	8468	2735	1927	884	115	0	0	0	34	280	873	1620		(13)
(14)	Wind	WSAvg	9.2	10.0	9.4	9.0	7.9	8.2	8.0	8.8	9.4	9.7	9.8	9.9	10.3	(14)
(15)	Precipitation	PrecAvg	13.3	1.1	0.7	0.9	1.0	1.3	1.1	1.3	1.5	1.4	1.3	1.0	0.7	(15)
(16)		PrecMax	24.9	4.4	3.7	5.0	3.3	3.7	3.5	3.9	3.5	3.4	4.6	2.8	4.2	(16)
(17)		PrecMin	5.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.1	0.0	0.1	0.0	(17)
(18)		PrecStd	4.4	1.1	0.9	1.1	0.8	0.9	0.7	0.8	0.9	0.9	1.2	0.8	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	109.4	106.2	100.3	88.8	76.5	67.1	66.2	73.4	84.4	94.2	100.9	104.5	(19)
(20)			MCWB	66.1	67.4	64.1	61.2	56.6	54.7	51.7	54.8	57.3	60.1	62.8	64.3	(20)
(21)		2%	DB	103.9	99.8	93.5	82.9	71.8	62.9	62.1	67.7	78.0	87.4	94.6	98.9	(21)
(22)			MCWB	66.0	66.4	64.1	59.5	56.5	52.7	51.2	52.7	55.8	58.6	62.4	63.5	(22)
(23)		5%	DB	98.4	95.0	89.0	79.0	67.9	60.4	59.3	63.9	72.8	82.1	89.5	94.2	(23)
(24)			MCWB	65.2	65.2	63.1	58.1	55.2	52.1	50.4	51.1	54.7	57.3	60.4	62.9	(24)
(25)		10%	DB	93.5	91.1	84.5	74.8	64.7	58.2	56.9	60.8	68.0	76.9	84.4	89.4	(25)
(26)			MCWB	64.9	64.6	62.3	57.4	54.2	51.1	49.4	50.3	53.3	56.2	59.9	61.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.1	74.3	69.7	64.4	61.8	58.1	55.7	57.4	61.7	65.1	69.2	71.3	(27)
(28)			MADB	83.9	87.8	85.0	73.7	66.4	61.6	60.9	66.0	71.9	78.8	83.2	83.3	(28)
(29)		2%	WB	70.9	71.7	67.6	62.3	58.6	55.1	53.2	54.7	58.8	62.1	66.4	67.8	(29)
(30)			MADB	85.7	85.7	82.1	74.3	66.3	59.8	58.3	63.4	70.3	76.3	80.5	81.8	(30)
(31)		5%	WB	69.0	69.7	65.8	60.7	56.9	53.5	51.7	52.7	56.7	60.0	64.1	65.6	(31)
(32)			MADB	87.0	83.8	81.3	74.0	64.9	58.2	56.8	61.6	68.4	74.5	80.3	85.0	(32)
(33)		10%	WB	67.3	67.6	63.9	58.9	55.3	52.1	50.4	51.1	54.7	57.9	62.0	63.7	(33)
(34)			MADB	87.3	84.8	80.6	72.7	63.2	57.1	55.5	59.2	66.2	74.0	79.1	84.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	28.1	26.3	25.1	24.3	20.2	17.4	17.3	20.5	24.2	27.0	27.1	28.3	(35)
(36)			MADB	34.2	31.5	30.7	30.2	24.5	20.4	20.3	26.1	31.2	35.3	33.9	33.8	(36)
(37)			MCWBR	10.2	9.5	10.4	12.6	12.0	11.6	11.3	13.0	13.4	13.2	10.9	10.4	(37)
(38)		5% WB	MADB	25.3	22.9	24.1	24.0	19.1	16.1	16.7	21.8	25.3	26.4	24.7	25.2	(38)
(39)			MCWBR	9.8	8.5	9.8	11.0	10.6	10.2	10.5	12.3	12.5	11.9	10.0	10.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.337	0.337	0.319	0.309	0.298	0.289	0.291	0.299	0.326	0.329	0.343	0.332	(40)	
(41)		taud	2.502	2.517	2.556	2.557	2.566	2.590	2.576	2.541	2.452	2.460	2.454	2.501	(41)	
(42)		Ebn at Noon	317	310	304	289	274	269	274	288	296	309	313	320	(42)	
(43)		Edh at Noon	36	34	30	27	23	21	23	27	33	36	37	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2497	2237	1831	1341	932	746	805	1115	1535	1991	2257	2503	(44)	
(45)		RadStd	114	104	86	62	47	56	37	69	108	130	121	104	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+283	+227
(47) Regional (12 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	-108	+263	+182

Nomenclature: See separate page